

IntroToArduino.odt

24may25

01jan26

Introduction to Using the Arduino Development System

ref1: <https://docs.arduino.cc/hardware/uno-rev3/>

IDEA: Make available project boards for the more complex examples.

Syllabus

0. Setting up the Arduino IDE

(see <https://www.arduino.cc/> > For Makers > Go to Documentation > Software > Arduino IDE 2)

1. Blink; the hardware “Hello World” application.
2. LEDs – Some electronics theory so that parts are not damaged.
3. Adding external LEDs to the Arduino aboard. Apply information from #2.

Experiments with six LEDs:

3a. Turn each LED ON; one at a time, then turn each LED OFF, one at a time.

3a1. Straight line code.

3a2. Using an **Array** to control which LED is Blinked.

3a3. Using an Array of **Data Structures** {LED#, State} and one function.

Use the Array as a circular pattern generator.

Use the Array as a scrolling patter generator; Chase, Ping-pong, Morse Code.

3a4. Using the random function to randomly pick an LED from the Array to turn ON/OFF or Toggle.

Use an Array to record the **State** of an LED.

3b. Use a **State Machine** to cycle through different display types.

3b1. Use time checks to set the run-time for each state.

3b2. Use a **Table** to control the state sequence.

4. 7 Segment LEDs

4a. Single 7-Segment LED

4a1. Using Arduino functions

4a2. Using direct hardware control registers

4b. Multiple 7-Segment LED